

Question

An alloy **M** is composed of metal **A** and metal **B**, and can form a variety of crystal structures such as X_1 -**M** phase, X_2 -**M** phase, X_3 -**M** phase, and X_4 -**M** phase. (To prevent directly searching for answers without reasoning, the real names of the structures are hidden)

X_1 -**M** phase is a phase formed when **B** is less, and its structure can be regarded as **B** atoms randomly replacing the positions of **A** atoms in metal **A**.

X_2 -**M** phase is a cubic crystal with a unit cell parameter of $a = 295$ pm; **A** atoms and **B** atoms are completely ordered in the crystal.

X_3 -**M** phase is also a cubic crystal, in which **A** atoms and **B** atoms are completely ordered, forming a four-layer nested atomic cluster of "**B** tetrahedron@**A** tetrahedron@**A** octahedron@**B** cuboctahedron"; the center of the atomic cluster is located at the vertices and body center of the unit cell (and the orientation of the atomic clusters at the vertices and body center are the same).

X_4 -**M** phase is a hexagonal crystal, and its structure can be regarded as **A** atoms randomly replacing the positions of **B** atoms in metal **B**.

Hint: (a) An ideal cuboctahedron is a highly symmetrical polyhedron surrounded by 6 square faces and 8 equilateral triangular faces; (b) The atomic radii of **A**, **B** are 128 pm, 133 pm, respectively.

The following statements are made:

1. The chemical formula of X_2 -**M** phase is A_3B , and **B** and **A** are located at the vertices and face centers of the unit cell, respectively.
2. The chemical formula of X_3 -**M** phase is AB .
3. If the fractional coordinate of an **A** atom in the "**A** tetrahedron" layer of the X_3 -**M** phase is (0.328, 0.672, 0.328), then the theoretical value of the unit cell parameter of the X_3 -**M** phase is $a = 891$ pm.

4. If the fractional coordinate of an **A** atom in the "**A** tetrahedron" layer of the $X_3\text{-M}$ phase is $(0.328, 0.672, 0.328)$, then the theoretical value of the unit cell parameter of the $X_3\text{-M}$ phase is $a = 744$ pm.

5. The lattice types of $X_2\text{-M}$ phase and $X_3\text{-M}$ phase are simple cubic and body-centered cubic, respectively.

What is the sum of the numbers of the correct statements?

A. 2

B. 3

C. 5

D. 7

E. 8

F. 10

G. 11

H. 12

I. All of the above options are incorrect.

Answer

Correct Answer: **E**

Detailed Explanation

Analyze X_2 -**M** phase: If **A, B** are in contact at the face diagonal, then $a = (128 + 133)\sqrt{2} = 369 \text{ pm} > 295 \text{ pm}$, which is inconsistent; if **A, B** are in contact at the body diagonal, then $a = (128 + 133) \times \frac{2}{\sqrt{3}} = 301 \text{ pm} \approx 295 \text{ pm}$, therefore **A, B** are in contact at the body diagonal, **A, B** are at the vertices and body center respectively, and the chemical formula is **AB**, statement 1 is incorrect.

CHECKPOINT

1 PTS

X_2 -**M** phase, **A, B** are in contact at the body diagonal, **A, B** are at the vertices and body center respectively, and the chemical formula is **AB**

Analyze X_3 -**M** phase: The composition of **B** tetrahedron@**A** tetrahedron@**A** octahedron@**B** cuboctahedron is **B₄@A₄@A₆@B₁₂**, the quadrilateral faces of the cuboctahedron are parallel to the faces of the cubic unit cell, and the two opposing triangular faces are parallel but oriented in opposite directions. Since the atomic clusters at the vertices and body center have the same orientation, the clusters at the vertices and body center do not share atoms, and each cluster is independent. The unit cell of the X_3 -**M** phase contains $2 \times (4 + 6) = 20$ **A** atoms and $2 \times (4 + 12) = 32$ **B** atoms, and the chemical formula is **A₅B₈**, statement 2 is incorrect.

CHECKPOINT

1 PTS

X_3 -**M** phase unit cell, the clusters at the vertices and body center do not share atoms

CHECKPOINT

1 PTS

$X_3\text{-M}$ phase chemical formula is A_5B_8

Calculate the unit cell parameter of $X_3\text{-M}$: Let the positional offset of the **A** atom relative to the vertex or body center be represented by the fractional coordinate t , then the coordinates of the **A** atoms in the "**A** tetrahedron" layer of the $X_3\text{-M}$ phase are $(t, t, t), (t, -t, -t), (-t, t, -t), (-t, -t, t)$ and the above four each plus $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, for a total of 8. It is easy to know that $(0.328, 0.672, 0.328) = (\frac{1}{2} - t, \frac{1}{2} + t, \frac{1}{2} - t)$, so $t = 0.172$.

CHECKPOINT

1 PTS

$t = 0.172$

Assume that the **B** atoms of the "**B** tetrahedron" are in close contact, and the length of the component of the distance from the **B** atom of the "**B** tetrahedron" to the center of the cluster in the direction of the coordinate axis is $133 \div \sqrt{2} = 94.0$ pm

CHECKPOINT

1 PTS

Length of the component of the distance from the **B** atom of the "**B** tetrahedron" to the center of the cluster in the direction of the coordinate axis is 94.0 pm

Assume that the **A** atoms of the "**A** tetrahedron" and the **B** atoms of the "**B** tetrahedron" are in close contact. Let the length of the component of the distance from the **A** atom of the "**A** tetrahedron" to the center of the cluster in the direction of the coordinate axis be w , according to the Pythagorean theorem, $\sqrt{(w - 94.0 \text{ pm})^2 + (w - 94.0 \text{ pm})^2 + (w + 94.0 \text{ pm})^2} = 128 \text{ pm} + 133 \text{ pm}$,

Solving for w : $w = 153.2 \text{ pm}$, unit cell parameter $a = 153.2 \text{ pm} \div 0.172 = 891 \text{ pm}$, statement 3 is correct, statement 4 is incorrect.

CHECKPOINT

1 PTS

$w = 153.2 \text{ pm}$

CHECKPOINT

1 PTS

$X_3\text{-M}$ phase unit cell parameter $a = 891 \text{ pm}$

The lattice types of the $X_2\text{-M}$ phase and the $X_3\text{-M}$ phase are simple cubic and body-centered cubic, respectively, statement 5 is correct.

CHECKPOINT

1 PTS

$X_2\text{-M}$ phase and $X_3\text{-M}$ phase lattice types are simple cubic and body-centered cubic, respectively

In summary, the correct statements are 3 and 5, so choose option E.